Topic 14: Switch Node & Workflow Routing

Project Overview

This project demonstrates how to use the Switch node in n8n to route multiple workflow items into different output branches based on user roles.

The workflow checks each user's role and directs the user to the appropriate downstream Edit Fields node. It also uses a fallback output to handle guest users and unmatched roles.

Objective

- Understand the Switch node in n8n.
- Configure multiple routing rules.
- Use expressions to evaluate incoming JSON data.
- Configure a fallback output.
- Process multiple items using the Split Out node.
- Connect separate downstream nodes to each output.

Workflow Structure

Manual Trigger → Mock Users → Split Users → Route by Role

The Switch node connects to four downstream branches:

1. Output 0 → Set Admin
2. Output 1 → Set Editor
3. Output 2 → Set Viewer
4. Output 3 → Set Guest (Fallback)

Sample Input Data

```
None
{
  "profiles": [
    {
```

```
      "name": "Alex",
      "role": "admin"
    },
    {
      "name": "Sarah",
      "role": "editor"
    },
    {
      "name": "John",
      "role": "viewer"
    },
    {
      "name": "David",
      "role": "guest"
    }
  ]
}
```

Routing Configuration

The Switch node uses the expression:

```
None
{{ $json.role }}
```

Routing rules:

Rule	Condition	Output
1	role equals admin	0
2	role equals editor	1
3	role equals viewer	2
Fallback	No matching rule	3

Downstream Actions

Node	Status Field	Value
Set Admin	status	Admin Access
Set Editor	status	Editor Access
Set Viewer	status	Viewer Access
Set Guest	status	Guest Access

The downstream nodes preserve the original input fields and add the corresponding status field.

Expected Results

- Alex is routed to Set Admin.
- Sarah is routed to Set Editor.
- John is routed to Set Viewer.
- David is routed to Set Guest.

How to Execute

1. Open the workflow in n8n.
2. Click Test workflow.
3. Wait for all nodes to execute.
4. Inspect the four downstream nodes.
5. Verify that each user reaches the correct branch.
6. Save the workflow.

Technologies Used

- n8n
- Manual Trigger
- Edit Fields (Set)
- Split Out
- Switch
- JSON
- Workflow Routing

Conclusion

The project demonstrates multi-branch workflow routing using the Switch node. It successfully processes multiple user profiles, evaluates role-based conditions, and directs users to the appropriate output, including a fallback route.